

GPS threats prompt air navigation innovation

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Navigating the complex world of EU mobility policy and infrastructure. By MARTINA SAPIO with TOMMASO LECCA SNEAK PEEK — Growing GPS jamming and spoofi

Navigating the complex world of EU mobility policy and infrastructure.

By MARTINA SAPIO

with TOMMASO LECCA

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SNEAK PEEK

— Growing GPS jamming and spoofing are pushing technology companies to develop new ways of keeping aircraft safely in the air when satellite navigation is disrupted.

— Yemen's Houthi rebels have attacked a commercial tanker in the Red Sea for the first time since September, just days after Iran warned the group could disrupt shipping through the Bab el-Mandeb Strait.

— Struggling German battery maker Varta is set to be broken up after creditors reportedly refused to provide fresh financing, marking another blow for a company once seen as a European battery champion.

Good morning, and welcome to Morning Mobility.

Tips to jdahl@politico.eu , tlecca@politico.eu and msapio@politico.eu . Follow us [@tommylecca](#) , [@jordyndahl](#) and [@martsapio](#) .

DRIVING THE DAY

FIA SEES INNOVATIVE SOLUTIONS FOR GPS JAMMING, SPOOFING: Since the start of Russia's full-scale invasion of Ukraine, airlines across Europe have increasingly faced GPS jamming — in which aircraft lose access to satellite navigation signals — and GPS spoofing, in which pilots are shown a false position.

GPS jamming and spoofing are typically linked to military activity that — intentionally or not — disrupts satellite signals across a given area, affecting aircraft, drones, ships and any other vehicle relying on GPS or other global navigation satellite systems. GPS jamming is now reportedly affecting around 1,000 civilian flights every day worldwide, mostly around conflict zones.

"You may be flying over Italy, but it's telling you that you're flying over St. Petersburg," said Aravind Ratnam, chief strategy officer at Q-CTRL, an Australian company developing a quantum navigation system designed to counter both jamming and spoofing. Along with several other companies active in the field, Q-CTRL showcased its technology at the Farnborough International Airshow.

EASA's eye on this: Earlier this month, the EU Aviation Safety Agency issued a revised safety information bulletin updating its recommendations to national authorities, air traffic controllers and airlines on the issue. The problem has existed since Feb. 2022, the agency said, but recent data "has shown further increase in the severity of its impact, as well as an overall growth of intensity and sophistication of these events." According to EASA, the areas most affected are "the south and eastern Mediterranean, Black Sea, Middle East, Baltic Sea, and Arctic area."

Thanks to ground-based navigation systems, which predate the introduction of satellite navigation for civil aviation, GPS interference rarely poses an immediate safety risk. Aircraft can generally continue flying and land safely even after losing satellite signals. However, EASA warned that the combination of multiple failures linked to jamming and spoofing "may have cumulative adverse effects on flight safety."

Airlines are already responding. Air France announced in May a multimillion-euro, three-year investment to equip its long-haul fleet with multi-mode receivers developed by Collins Aerospace, part of RTX. The equipment combines multiple navigation sources, allowing aircraft to continue navigating even if one system becomes unavailable.

Back to nature: But according to Ratnam, even "the most expensive systems in the world can give you one mile of inaccuracy for every hour of flying," because of limitations that have nothing to do with the receiver itself. That is why Q-CTRL believes it has a better solution: using the Earth's magnetic field for navigation.

Very down to earth: "The Earth has two sources of magnetism, one that comes from the center of the Earth, and one that come from near the surface," Ratnam told Tommaso. "Near the surface of the Earth, there are mineral deposits and rock formations that cause another weaker magnetic signal. What we are picking up is that," he added.

Mapping magnetism: Q-CTRL first creates a magnetic map by flying a magnetometer over the flight path it wants to protect from jamming. Once the map has been established, the system can determine an aircraft's position by matching the magnetic signals it detects with those on the map. The system relies on magnetic signatures near the Earth's surface rather than the planet's core because the latter field "slowly changes over time, whereas the magnetic field that's near the surface will stay stable for millions of years."

Going quantum: The magnetometer "uses atomic cell that interacts with the magnetic field and it exhibits certain quantum-like interactions," Ratnam said, which is why the company describes its technology as a quantum navigation solution.

Not yet ready: Asked when the technology could become available to airlines, Ratnam said Q-CTRL is backed by Airbus and Lockheed Martin, which are helping the company through the certification process. The system is now ready for flight testing and "We are moving to technology readiness level 9 hopefully not too long from now," he added, referring to the highest level on the technology readiness scale .

MARITIME

HOUTHIS MAKE GOOD ON RED SEA THREAT: Yemen's Houthis have attacked a commercial tanker in the Red Sea for the first time since September, escalating fears the group is following through on Iran's warning that it could disrupt shipping through the Bab el-Mandeb Strait. The U.K. Maritime Trade Operations said the crew of one tanker was battling a fire after it was struck by an unidentified projectile.

Conflicting claims: The Houthis said they targeted two tankers, although only one attack has so far been corroborated by UKMTO. Iran also claimed to have stopped three tankers attempting to transit the Strait of Hormuz, with one reportedly hit and two turning back.

POLITICO contacted the Houthis for comment but did not receive an immediate response.

Blockade taking shape: The attack comes days after the Houthis declared a “complete naval blockade” of Saudi Arabia, opening a new front in the U.S.-Iran conflict. While the group has not explained how it intends to enforce the blockade, the Bab el-Mandeb Strait is the main gateway to the Red Sea and a critical route for Saudi oil exports. Reuters reported last week that Iran had instructed the Houthis to stand ready to disrupt shipping through the chokepoint if the U.S. struck Iranian power infrastructure.

Selective access? Even as they threaten commercial shipping, the Houthis appear willing to grant passage to some vessels. Chinese state-owned shipping giant Cosco is expected to send one of its tankers through Bab el-Mandeb after twice turning around in the Red Sea, suggesting transits may now be approved on a case-by-case basis.

Why it matters: A full closure of Bab el-Mandeb would be more difficult than Iran’s blockade of the Strait of Hormuz because vessels can still access the Red Sea via the Suez Canal. But prolonged disruption of both chokepoints would threaten a significant share of global oil flows, drive up freight and energy costs, and further strain already fragile supply chains. The latest developments suggest the Houthis are moving beyond sporadic attacks toward exercising selective control over one of the world’s busiest shipping lanes.

NEW SANCTIONS PACKAGE TOOTHLESS, ANALYSTS SAY: The EU’s new sanctions package will do little to target Russia’s lucrative liquefied natural gas trade after member countries decided to allow Greek shipping giant Dynagas to continue transporting vast quantities of the Russian product indefinitely, sanctions analysts told our colleague Ben Munster.

The compromise: Under the newly compromised 21st sanctions package, shipping companies transporting Russian LNG outside the EU under contracts signed before the invasion of Ukraine will be able to continue operating for a 12-month period subject to renewal, two EU diplomats told my colleague Koen Verhelst, speaking on condition of anonymity.

Heavy roller: The compromise was reached after Athens complained that banning such existing contracts outright would jeopardize the lucrative Russia trade enjoyed by Greek tanker companies — principally Dynagas, a shipping giant owned by Greek billionaire George Prokopiou, whose fleet of specialized icebreakers has become critical to Russia’s LNG export business.

Too little, too late: While the ban prevents shipping firms from cutting future deals with Novatek, analysts say the exemption will preserve the entirety of the Greek shipping industry’s transport contracts with Russia, which were established well before the war in 2015.

Lion’s share: Dynagas, for instance, already facilitates a third of Novatek’s LNG export business, accounting for 35 percent of all exports from its Yamal LNG megaproject in the first half of this year and operating five of the fifteen specialized Arc7 tankers Yamal needs to transit icebound Arctic seas, according to Alexander Kirk, a top sanctions campaigner at NGO Urgewald.

Poor form: If the measures are confirmed, “this would significantly dilute or postpone the transport measure’s impact on Dynagas’ Yamal LNG operations,” he said, adding that his assessment may still

change when the full legal text of the sanction agreement is made public.

CARS

VARTA RUNS OUT OF JUICE: Struggling German battery-maker Varta is set to be broken up after creditors reportedly refused to provide fresh financing, marking another blow for a company once seen as a European battery champion.

From darling to distress: Varta built its reputation supplying lithium-ion cells for Apple's AirPods and producing consumer and automotive batteries, but failed to establish itself as a major EV battery player despite investing in new cell technology and partnering with Porsche. After soaring during the battery boom, its shares collapsed as the company underwent restructuring between 2022 and 2024.

The endgame: After Apple ended its battery purchases and Varta announced 350 job cuts at its Bavarian plant in May, the company's main creditors reportedly rejected further funding at a meeting on Wednesday. According to a report by the Frankfurter Allgemeine Zeitung, the household battery division will now be carved out as creditors seek a new owner.

Headlines

Here's a recap of yesterday's news, along with Pro articles and alerts from overnight.

Russia chokes Ukraine's lifeline Black Sea export corridor

Moscow "is holding global food security hostage," Ukrainian Foreign Minister Andrii Sybiha said on Wednesday.

By Veronika Melkozerova · Jul 23, 2026, 12:17 pm Europe/Brussels CEST

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